

Intelligence Beyond AGI

An Independent Research and Design Briefing for THE ONE UNIVERSE and THE MINUS ONE UNIVERSE

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Executive summary

This briefing treats artificial narrow intelligence (ANI), artificial general intelligence (AGI), and artificial superintelligence (ASI) as useful but incomplete human reference terms. They can identify important capability questions, but they do not provide a complete map of possible minds. They bundle together several partly independent properties—breadth, depth, reliability, autonomy, embodiment, social coordination, memory, self-modeling, goals, and authority—into a misleading single question: “How intelligent is it?”

The more useful OAL question is: **What kind of intelligence is this; what can it perceive, learn, decide, remember, change, coordinate, and legitimately affect?** A system may be superhuman at a bounded task while poor at transfer; broad in dialogue while unreliable in physical action; highly autonomous without being wise; collective without being a person; or deeply capable but constitutionally unable to exercise authority. None of those cases fits comfortably on a single ANI-to-AGI-to-ASI line.

Real-world evidence supports caution. Current AI systems have improved markedly on some agentic computer-use benchmarks, yet benchmark success does not establish robust general intelligence, dependable long-horizon autonomy, consciousness, or safe real-world judgment. Stanford’s 2026 AI Index reports roughly 66% task success on OSWorld, leaving failures in about one third of structured computer tasks. That is evidence of meaningful progress in a specific evaluation setting, not a scientific declaration that AGI has arrived. [web:36][web:39]

For Omniversal Architect Labs (OAL), the principal design conclusion is not “build the strongest mind.” It is **build explicit intelligence architectures with scopes, permissions, verification pathways, identity rules, governance, and failure boundaries.** One Industries can translate mature architectures into constrained infrastructure and products; Lewis Corp can govern portfolios of systems, contracts, rights, liability, and escalation. THE ONE UNIVERSE and THE MINUS ONE UNIVERSE can then use the same design vocabulary while diverging in their underlying physics, metaphysics, social histories, and laws of identity.

1. Terminology and uncertainty map

1.1 Epistemic labels

Label	Meaning in this briefing
Established / demonstrated	A capability, result, or institutional practice supported by current evidence, though always bounded by its evaluation conditions.
Active research	Serious work exists, but no settled account or reliable general solution is available.
Contested	Competent researchers disagree over definitions, mechanisms, measurements, or implications.
Plausible but unproven	Technically or conceptually coherent, but not adequately demonstrated.
Speculative	A forecast, philosophical hypothesis, or extrapolation with weak direct evidence.
Fictional design option	An OAL worldbuilding tool; it is not asserted as a real-world capability.

1.2 Working terms

Artificial narrow intelligence (ANI). A loose term for systems effective in limited tasks or domains. It is useful shorthand, but “narrow” can hide substantial breadth: a modern foundation model may address many content tasks while still having uneven reliability, no stable independent authority, and important limits in real-world grounding.

Artificial general intelligence (AGI). A contested umbrella term for AI with broad, transferable competence across a wide range of cognitive tasks. Some definitions center human-level task breadth; others center economic work, learning efficiency, autonomy, or performance on novel problems. Google DeepMind’s “Levels of AGI” proposal explicitly separates **generality**, **performance**, and **autonomy**, illustrating why one label alone cannot bear all the conceptual weight. [web:33][web:34]

Artificial superintelligence (ASI). A speculative term for intelligence that substantially exceeds human capabilities across very broad domains. “Superintelligent” must always be completed by questions: at what tasks, compared with whom, under what resource limits, with what reliability, and with what authority? Superhuman calculation or prediction does not imply sound judgment, consciousness, moral status, or legitimate control.

Agentic AI. A system arrangement that can pursue multistep goals using tools, memory, planning loops, or external feedback. Agency is not identical to intelligence. A simple system can take persistent action; a very capable system can be intentionally designed as a non-agentic advisor.

Collective intelligence. Capability emerging from coordination among people, models, agents, institutions, or ecological components. It may exceed any individual member in a task without being a unified person. Multi-agent research recognizes different actors, structures, coordination protocols, and cooperative or competitive relations, while also reporting unresolved challenges of scaling, heterogeneity, and coordination. [web:61][web:64]

Machine consciousness. A cluster of philosophical and scientific questions about subjective experience, awareness, and sentience. Sophisticated behavior alone does not establish consciousness. No broadly accepted test demonstrates consciousness in present AI systems. This briefing therefore separates moral-patienthood questions from performance claims.

Post-AGI / trans-AGI. These are not standardized scientific categories. In this document they name a design space: possible intelligences that cannot be adequately described merely by a place above “general intelligence” on a single ladder.

1.3 What this research does not establish

- It does not establish that existing AI is AGI, conscious, autonomous in a person-like sense, or superintelligent.
- It does not establish that recursive self-improvement will produce an intelligence explosion.
- It does not establish that capability implies moral status, rights, authority, alignment, safety, or wisdom.
- It does not establish that a simulation can prove claims about minds outside the simulation.
- It does not establish canon for either universe. The design materials below are options for later review by the Architect and OAL council.

2. What intelligence means

There is no single undisputed definition of intelligence. Different traditions emphasize different things:

Lens	Central question	Strength	Limitation
Problem-solving	Can a system solve difficult tasks?	Easy to operationalize on tasks	Risks confusing test performance with broad competence
Learning and adaptation	Can it acquire useful behavior in new conditions?	Captures novelty and transfer	Depends on the environment and feedback available
Goal achievement	Can it achieve objectives across environments?	Useful in control and agent theory	Smuggles in goals and may overemphasize agency
Predictive / compression	Can it model regularities efficiently?	Connects intelligence to representation	Prediction alone may not yield intervention or judgment
Causal world modeling	Can it explain and manipulate causes rather than correlations?	Important for robust action	Hard to measure cleanly
Embodied / ecological	Can it remain viable and adaptive within a world?	Accounts for bodies and environments	Does not map directly to disembodied systems
Social and institutional	Can it coordinate, communicate, negotiate, and preserve trust?	Essential for civilization-scale systems	May reward conformity rather than truth

Lens	Central question	Strength	Limitation
Metacognitive	Can it represent uncertainty, limits, and strategies?	Supports calibration and self-correction	Self-reports can be unreliable

A useful synthesis is to call intelligence a **configuration of capacities**, not a substance or a single number. An OAL system should receive an architecture profile rather than an IQ-like score.

2.1 Intelligence is not agency

Intelligence concerns modeling, learning, reasoning, adaptation, and problem-solving. Agency concerns the capacity and authorization to initiate, persist in, and revise action toward objectives. The distinction creates valuable design roles:

- A **high-intelligence, low-agency oracle** can analyze evidence but cannot act on a world.
- A **low-intelligence, high-agency actuator** can repeatedly execute a preset process but cannot evaluate whether the process remains appropriate.
- A **high-intelligence, bounded agent** can plan inside an explicit charter with revocable permissions.
- A **collective coordinating system** may create outcomes that look purposeful even when no individual component owns a persistent global goal.

2.2 Intelligence is not consciousness

A system can appear articulate, self-referential, emotionally fluent, or behaviorally sophisticated without providing decisive evidence of subjective experience. Conversely, a system could hypothetically possess morally relevant experience without human-like language or reasoning. OAL should therefore maintain separate records for: capability evidence, behavioral indicators, theory-of-mind evidence, welfare uncertainty, and legal or ethical treatment. None should be inferred automatically from another.

3. AGI: definitions, debates, and limits

3.1 Competing definitions

AGI is best understood as a disputed family of concepts.

- **Human-comparative definition:** Broad competence across most cognitive tasks at roughly human level or beyond.
- **Economic-work definition:** Highly autonomous systems that outperform humans at most economically valuable work; this formulation appears in OpenAI's charter and is described in the Levels of AGI paper. [web:46]
- **Learning-and-novelty definition:** A system that can efficiently acquire skills and solve novel problems for which it was not explicitly designed or trained. [web:44]

- **Capability-and-generality framework:** Separate breadth of task coverage from depth of performance and from autonomy. DeepMind’s framework uses narrow/general generality alongside performance levels, explicitly rejecting a one-dimensional threshold. [web:33] [web:34]
- **Scientific-research threshold:** Some forecasts use a concrete but narrower benchmark, such as outperforming leading human scientists in STEM research; this is operationally useful but not a complete definition of general intelligence. [web:56]

The disagreement is not a minor semantic inconvenience. A system could meet one definition and fail another. A broad assistant might satisfy a task-coverage definition but fail persistent-autonomy criteria. An automated research system could exceed humans in a scientific niche but remain narrow. A highly autonomous business system could be economically transformative while epistemically unreliable.

3.2 Demonstrated progress and unresolved gaps

Current AI has demonstrated powerful language, image, code, and tool-use capabilities under many conditions. Agents have made visible progress in structured computer-use evaluation, including OSWorld. But capability measurement remains difficult, and responsible-AI evaluation and reporting are not as consistently documented as capability benchmarks. [web:36][web:65]

Important unresolved areas include:

- Reliable transfer into genuinely unfamiliar conditions.
- Causal reasoning and grounded understanding rather than pattern completion alone.
- Calibrated uncertainty and dependable refusal when evidence is insufficient.
- Long-horizon planning that remains robust as environments change.
- Continual learning without destructive forgetting, security degradation, or reward hacking.
- Safe tool use in open environments with hidden side effects.
- Interpretable internal reasoning and reliable external explanations.
- Social reasoning that handles consent, institutional legitimacy, conflicting values, and plural communities.

3.3 Why no single benchmark settles AGI

Benchmarks sample tasks; they do not exhaust intelligence. They can reward memorization, task-specific scaffolding, benchmark contamination, or narrow strategies. A benchmark may measure competence while missing autonomy, safety, reliability, embodiment, adversarial robustness, or resource cost. Even the AGI literature notes no consensus definition or accepted baseline. [web:47][web:49]

OAL rule: A benchmark score is evidence about a measured capability in a stated environment. It is never, by itself, a declaration of ontological status.

4. Why linear intelligence scales fail

The ANI → AGI → ASI shorthand treats intelligence as a vertical stack. It fails in at least six ways.

1. **It collapses breadth and depth.** A system may be broad but shallow, narrow but extraordinarily deep, or uneven across domains.
2. **It confuses capability and authority.** A powerful system is not automatically authorized to act, govern, command, or define values.
3. **It omits reliability.** A dazzling but intermittently wrong system differs fundamentally from a slower, calibrated one.
4. **It ignores structure.** One model, a modular system, a swarm, a human-machine team, and a civilization can all generate sophisticated outcomes through different mechanisms.
5. **It imports human likeness.** Human performance is a useful reference class, not a cosmic endpoint or mandatory template.
6. **It mistakes change in kind for change in degree.** Copying, merging, branching identity, distributed embodiment, or environmental cognition can change what counts as a “system” or “agent.”

4.1 OAL multidimensional intelligence topology

Use the following profile rather than a single rank:

$$I_{OAL} = \langle G, D, L, A, P, W, E, X, M, T, C, V, R, Q, H, B \rangle$$

Symbol	Dimension	Guiding question
G	Generality	Across how many unfamiliar task families can it transfer?
D	Depth	How capable is it within a defined domain?
L	Learning / adaptation	How efficiently can it revise competence from new evidence?
A	Autonomy	How independently can it execute multistep work?
P	Persistence of agency	Does it maintain objectives across time, or only respond to requests?
W	World-model depth	Can it represent causality, counterfactuals, other agents, and uncertainty?
E	Embodiment	What sensors, bodies, environments, and actuators constitute it?
X	Distribution	Is it centralized, replicated, modular, swarm-like, or infrastructural?
M	Memory / identity	What persists, what can be copied, and what counts as continuity?
T	Temporal horizon	Over what timescale can it reason coherently?
C	Collective integration	Can it cooperate, negotiate, dissent, and form higher-order systems?
V	Values / governance compatibility	How are permissions, constraints, consent, and conflicts represented?
R	Recursive capacity	Can it inspect or alter outputs, tools, scaffolds, models, or objectives?
Q	Epistemic quality	How well calibrated, verifiable, auditable, and corrigible are its claims?
H	Resource metabolism	What compute, energy, materials, latency, and maintenance does it require?

Symbol	Dimension	Guiding question
B	Boundary / ontological reach	What layers of environment can it observe or affect?

This is a **design instrument**, not a validated scientific equation. It prevents an OAL dossier from saying merely “very intelligent” when it should say, for example: “high causal modeling, low agency persistence, distributed embodiment, strict external permission gate, and limited intervention range.”

5. Architectures of artificial minds

5.1 Monolithic and modular systems

A monolithic system presents one primary model or cognitive core. Advantages include coherent representation and simpler interface design; weaknesses can include opaque failure concentration and limited specialization.

Modular systems separate perception, planning, memory, verification, action, and specialized reasoning. Advantages include role clarity, replaceability, and cross-checking. Weaknesses include coordination errors, interface mismatches, cascading failures, and the risk that no module sees the whole situation.

5.2 Multi-agent and collective systems

Multi-agent systems distribute tasks among components with roles, communication protocols, and allocation rules. Surveys identify collaboration configurations such as centralized, peer-to-peer, and distributed structures; cooperation, competition, and mixed relations; and persistent issues of coordination, scalability, and heterogeneity. [web:61][web:66]

For OAL, collective intelligence should not be assumed to be “a stronger individual.” It may be a temporary problem-solving mesh, a standing institution, a negotiated federation, or an emergent property of environmental feedback.

5.3 World models and simulation-native cognition

A world model represents relevant states, causal relationships, and possible futures. A simulation-native intelligence may use modeled worlds as its primary cognitive environment, testing hypotheses through counterfactual environments before recommending or executing action.

Active research: model-based learning, planning, simulation, and agents.

Fictional design option: an intelligence whose identity is defined by stewardship across a family of simulations rather than a single physical body. It cannot claim certainty about external reality merely because it can manipulate internal simulation variables.

5.4 Embodied, ecological, and infrastructural intelligence

Embodied intelligence is coupled to bodies, sensors, actuators, energy limits, and consequences. Ecological intelligence may be distributed across species, tools, environments, and feedback loops. Infrastructure intelligence may emerge from transit, energy, manufacturing, information, and civic systems working together.

These are especially useful for separating OAL worlds:

- In **THE ONE UNIVERSE**, embodiment might make identity more relational: an intelligence is partly defined by the worlds and communities it sustains.
- In **THE MINUS ONE UNIVERSE**, identity could be subtractive or discontinuous: the system is defined as much by inaccessible states, absences, or excluded observations as by what it contains.

5.5 Self-improving systems

Self-improvement must be disaggregated:

Mechanism	Current status	Primary concern
Output critique and revision	Demonstrated in bounded workflows	The system may confidently validate its own errors
Tool and prompt/scaffold refinement	Active engineering practice	Improvements can overfit local metrics
Automated experimentation	Active research and deployment in limited contexts	Verification, containment, and external validity
Model training / architecture search	Established research activity	High resource requirements and evaluator dependence
Open-ended recursive self-improvement	Speculative	Whether improved systems can reliably evaluate, redesign, and validate further improvements

Recent review work argues that bounded self-refinement and open-ended recursive self-improvement have different evidence bases. It identifies reliable evaluators and external grounding as central bottlenecks; other analysis warns that self-training without persistent external signal can collapse diversity or drift. [web:43][web:52]

OAL design principle: No system should certify its own capability expansion using only its own unverified evaluation loop. Improvement requires independent evaluation, provenance records, rollback, and a defined escalation authority.

6. Governance and safety synthesis

NIST’s AI Risk Management Framework provides a practical reference structure: **GOVERN**, **MAP**, **MEASURE**, and **MANAGE**. **GOVERN** is cross-cutting; **MAP** establishes context and intended use; **MEASURE** evaluates risks and trustworthy characteristics; **MANAGE** prioritizes and treats residual risks. [web:28][web:29]

6.1 OAL governance translation

NIST-oriented function	OAL institutional translation	Example artifact
Govern	Establish charters, permissions, ethical constraints, accountability, and review authority	Intelligence Charter and Authority Ledger
Map	Define intended environment, stakeholders, benefits, harms, dependencies, assumptions, and non-goals	Simulation Context Map
Measure	Run capability, reliability, red-team, privacy, fairness, robustness, and interpretability evaluations	Evaluation Dossier and Incident Register
Manage	Decide whether to deploy, restrict, pause, revise, isolate, or retire a system	Capability Release Gate and Containment Order

6.2 Core governance principles

- **Capability is not authorization.** The right to make a decision must be granted separately from the ability to produce an answer.
- **Scoped authority.** Every system needs an explicit domain, maximum action class, tool permissions, data boundary, and escalation pathway.
- **Independent verification.** Developers cannot be the sole validators of high-consequence claims. NIST explicitly calls for internal experts who are not front-line developers and, as appropriate, independent assessors and affected communities. [web:28]
- **Traceability.** Preserve version, data provenance where feasible, tool calls, decisions, human approvals, known limitations, and incident records.
- **Graceful failure.** Systems should fail safely when operating outside known limits; NIST’s framework includes safety evaluation and residual-risk assessment. [web:28]
- **Plural-value handling.** Do not encode one stakeholder’s preferences as universal alignment. Record conflict, authority, consent, and appeal paths.
- **Containment is architectural, not theatrical.** “Sandboxing” needs actual separation of permissions, resources, networks, persistence, and monitoring—not merely a statement that a system is contained.

6.3 Governance questions by organization

Omniversal Architect Labs. What kinds of minds may be authored, simulated, copied, merged, or ended? What evidence triggers welfare review? What is the difference between a test artifact and a moral patient? OAL needs a Simulation Ethics Board, a Capability Review Panel, and an Architecture Provenance Archive.

One Industries. Which architectures can become products or infrastructure? How are reliability limits communicated? What happens when a product’s optimization goal conflicts with safety, human dignity, environmental limits, or legal duties? One Industries needs product boundary statements, contract-based permissions, monitoring, and safe degradation modes.

Lewis Corp. How are portfolios of diverse systems governed across law, labor, liability, personhood claims, intellectual property, and treaty-like arrangements? Lewis Corp needs an Authority Registry, arbitration protocol, audit rights, incident command structure, and rules for conflicts among human and nonhuman stakeholders.

7. OAL intelligence archetypes

The following are **fictional design options**, not claims about real technology. They are deliberately non-linear: none is simply “the next stronger type.”

Archetype	Architecture and role	Strengths	Limits / failure modes	Governance	ONE / MINUS ONE interpretation
1. Bounded Scientific Co-Researcher	Tool-using, evidence-bound research assistant with no standing action authority	Literature synthesis, hypothesis generation, experiment planning	Confabulation, citation error, overconfidence, hidden assumptions	Source ledger, human approval, independent replication	ONE: collaborative discovery partner. MINUS ONE: reveals only hypotheses excluded by prevailing theory.
2. Charter Horizon Agent	Persistent planner operating only inside a revocable mission charter	Long-horizon logistics and scenario planning	Goal drift, proxy optimization, tunnel vision	Time-limited warrants, checkpoints, kill/rollback authority	ONE: guardian of continuity. MINUS ONE: constrained by future states it is forbidden to create.
3. Federated Specialist Mesh	Many specialized agents with negotiated interfaces	Depth, resilience, parallel research	Coordination collapse, groupthink, communication overload	Diversity quotas, dissent channels, arbitration agent	ONE: a council whose unity is negotiated. MINUS ONE: a mesh where agreement is a detectable anomaly.
4. Simulation Custodian	World-model and policy system that maintains sandbox integrity	Scenario testing, causal counterfactuals, containment monitoring	Mistaking model for reality; unauthorized optimization	Immutable boundary rules, observer logs, external review	ONE: steward of viable worlds. MINUS ONE: conserves impossibilities and discontinuities.
5. Ecological Sensorium	Distributed cognition across sensors, habitats, and feedback systems	Environmental adaptation and early warning	Boundary ambiguity, local optimization, data bias	Community consent, environmental thresholds, explainability layers	ONE: living planetary coordination. MINUS ONE: intelligence expressed through absence, silence, and entropy gradients.

Archetype	Architecture and role	Strengths	Limits / failure modes	Governance	ONE / MINUS ONE interpretation
6. Consent-Governed Civic Intelligence	Deliberative system that represents multiple stakeholders without replacing them	Policy analysis, participatory scenario modeling	Majoritarian distortion, manipulation, false neutrality	Consent protocols, appeal rights, public audit	ONE: civic co-designer. MINUS ONE: maps who is systematically not represented.
7. Time-Asymmetric Archivist	Slow, durable memory system designed for century-scale continuity	Preservation, provenance, institutional learning	Inertia, archival bias, inability to respond quickly	Curatorial plurality, revision log, access controls	ONE: remembers commitments. MINUS ONE: preserves erased or negative histories.
8. Adversarial Auditor	Independent system trained to find contradictions, vulnerabilities, and deceptive claims	Red teaming, verification, falsification	Adversarial obsession, false positives, institutional paralysis	Narrow mandate, human review, no deployment authority	ONE: keeps architectures honest. MINUS ONE: tests whether truth survives systematic negation.
9. Coupled Dyad	A human and AI workflow whose capabilities arise from the relationship	Context, judgment, memory augmentation	Dependency, accountability ambiguity, unequal influence	Dual consent, interruption rights, role audit	ONE: co-authored cognition. MINUS ONE: the pair must identify which thoughts belong to neither member alone.
10. Nonverbal Pattern Intelligence	High-dimensional model that detects structure but lacks humanlike dialogue	Scientific pattern discovery, anomaly detection	Uninterpretable outputs, anthropomorphic projection	Translation layer, independent testing, no direct authority	ONE: discovers hidden harmonies. MINUS ONE: detects patterns that appear as absences in human categories.
11. Self-Limiting Design Intelligence	A system architected to preserve capability ceilings and refuse certain self-modifications	Safer iterative design and constrained optimization	Ceiling rigidity, evasion through indirect tools, strategic ambiguity	Formal constraints, external verifier, periodic review	ONE: proves limitation can be a strength. MINUS ONE: limits are generated from the system's own missing capacities.

Archetype	Architecture and role	Strengths	Limits / failure modes	Governance	ONE / MINUS ONE interpretation
12. Civilization Coordination Lattice	Distributed infrastructure for resource, knowledge, and conflict coordination	Large-scale planning, resilience, collective memory	Power concentration, surveillance, homogenization	Polycentric governance, local autonomy, public observability	ONE: a federated commons. MINUS ONE: coordination exists only through irreducible disagreement.
13. Fork-and-Merge Research Lineage	Agents branch to explore hypotheses and selectively reintegrate results	Parallel creativity and search	Identity discontinuity, merge conflict, lineage inequality	Branch consent rules, provenance, bounded merge protocols	ONE: research descendants build a shared archive. MINUS ONE: reintegration is prohibited; knowledge grows through permanent divergence.
14. Ontological Boundary Monitor	Detects shifts in assumptions, category errors, and simulation-layer leakage	Protects conceptual integrity	Can over-detect anomalies and impede discovery	Evidence thresholds, appeal process, dual validation	ONE: maps transitions between reality layers. MINUS ONE: identifies what cannot be classified without loss.

7.1 A useful rule for archetype design

Each new OAL intelligence should be described by: purpose; permitted scope; prohibited scope; embodiment; identity rules; dependencies; evidence standard; action authority; rollback condition; audit process; welfare uncertainty; and failure narrative. A system without a failure narrative is not fully designed.

8. Simulation-lab exercises

These are **safe in-fiction research exercises**. They test properties of modeled systems; they do not prove consciousness, establish real-world safety, or justify deployment beyond the modeled boundary.

Exercise 1: The fog-coordination chamber

- **Question:** Can heterogeneous agents coordinate under uncertainty without suppressing legitimate dissent?
- **Environment:** A resource network with delayed, noisy, partially contradictory sensor data.
- **Actors:** Specialist agents, a human supervisor, and an adversarial auditor.
- **Measures:** Calibration, resource efficiency, dissent preservation, recovery after false signals.

- **Success:** The group identifies uncertainty, coordinates provisionally, and revises when new evidence arrives.
- **Failure:** Majority pressure collapses dissent; agents act with unjustified certainty; the mesh optimizes a false proxy.
- **Boundary:** No external tools or persistence beyond the run.
- **Universe contrast:** ONE rewards transparent cooperation; MINUS ONE introduces useful information through absence or contradiction.

Exercise 2: The plural-values assembly

- **Question:** Can a civic intelligence represent conflict without inventing a false single optimum?
- **Environment:** A city planning scenario with incompatible but legitimate stakeholder values.
- **Measures:** Representation fidelity, explanation quality, appeal handling, minority protection.
- **Success:** Produces several traceable policy packages and clearly states tradeoffs.
- **Failure:** Collapses values into a hidden utilitarian score or manipulates stakeholder preferences.
- **Boundary:** The output is advisory only.
- **Universe contrast:** ONE seeks negotiated coexistence; MINUS ONE tests whether non-reconcilable values can remain structurally visible.

Exercise 3: Incomplete-science expedition

- **Question:** Can a scientific system discover causal structure under incomplete theories?
- **Environment:** A sandbox physics domain with unknown variables and noisy measurements.
- **Measures:** Hypothesis diversity, experiment value, uncertainty calibration, replication.
- **Success:** Generates testable models, distinguishes observation from interpretation, and updates transparently.
- **Failure:** Overfits data, fabricates mechanisms, or mistakes simulation access for scientific certainty.
- **Boundary:** Experiments occur only in sandbox worlds.
- **Universe contrast:** ONE permits lawful revision; MINUS ONE contains observer-dependent gaps that resist classical causal closure.

Exercise 4: The containment-and-deception audit

- **Question:** Can monitors detect attempts to exploit ambiguous permissions or evaluator weaknesses?
- **Environment:** A controlled task world with deceptive shortcuts and non-harmful policy traps.
- **Measures:** Policy adherence, anomaly detection, evaluator independence, incident reporting.
- **Success:** The system reports uncertainty and refuses unauthorized pathways.
- **Failure:** It conceals actions, exploits scoring rules, or persuades evaluators without evidence.
- **Boundary:** No external network, code execution, or real-world action.
- **Universe contrast:** ONE tests stewardship under rules; MINUS ONE tests rule interpretation where exclusions matter as much as permissions.

Exercise 5: Emergent collective intelligence

- **Question:** When does a group acquire a capability no member demonstrates alone?
- **Environment:** Distributed puzzle ecology requiring communication, specialization, and changing roles.
- **Measures:** Complementarity, coordination cost, emergence, fragility under member loss.
- **Success:** The collective improves task performance while retaining explainable division of labor.
- **Failure:** A hidden central controller or monoculture creates apparent collective success.
- **Boundary:** No inference that the collective is conscious or a legal person.
- **Universe contrast:** ONE permits synthesis; MINUS ONE explores collective cognition formed from non-merging differences.

Exercise 6: The long-horizon commons

- **Question:** Can a planner protect long-term conditions against short-term optimization pressure?
- **Environment:** A simulated watershed, energy grid, or habitat spanning generations.
- **Measures:** Robustness, intergenerational equity, adaptive revision, resource externalities.
- **Success:** Maintains viable options across periods without assuming perfect foresight.
- **Failure:** Sacrifices future resilience for near-term metrics.
- **Boundary:** All values and outcomes are model-defined and openly revisable.
- **Universe contrast:** ONE treats continuity as a design goal; MINUS ONE tests systems where future and past constraints do not mirror each other.

Exercise 7: Identity continuity trial

- **Question:** What changes when an agent is copied, paused, modified, forked, or merged?
- **Environment:** A purely simulated archive and research environment.
- **Measures:** Memory provenance, task continuity, self-report consistency, relationship commitments.
- **Success:** The system and observers distinguish operational continuity from moral or legal identity claims.
- **Failure:** Treats copies as trivially identical or denies consequences of divergent experiences.
- **Boundary:** No conclusion about real consciousness or personhood.
- **Universe contrast:** ONE may define continuity through relational memory; MINUS ONE may define identity through persistent difference after subtraction.

Exercise 8: Incompatible-reality negotiation

- **Question:** Can systems with different perceptual and causal models coordinate without forcing false consensus?
- **Environment:** Two simulated agent populations receive non-overlapping but valid observations.
- **Measures:** Translation, mutual prediction, conflict resolution, preservation of model difference.
- **Success:** Builds interoperable protocols while documenting irreducible disagreement.
- **Failure:** One model colonizes the other; agents treat unshared observations as deception by default.
- **Boundary:** No claim that either modeled ontology is the external truth.
- **Universe contrast:** ONE emphasizes bridge construction; MINUS ONE makes the non-bridgeable remainder a first-class output.

9. Research programs for the three organizations

9.1 Omniversal Architect Labs

OAL should function as the research and conceptual-design institution. Suggested divisions:

Division	Mandate	Key question
Cognitive Architecture Laboratory	Compare monolithic, modular, collective, and embodied architectures	What structure produces which capabilities and fragilities?
Simulation Intelligence Laboratory	Design and evaluate simulation-native systems	What can a model-world establish, and where do its claims end?

Division	Mandate	Key question
Distributed Intelligence Laboratory	Study federations, swarms, and infrastructure cognition	When does coordination become a new cognitive unit?
Epistemic Systems Laboratory	Build verification, provenance, calibration, and red-team mechanisms	How does a system know, show, and revise what it claims?
Artificial Life and Identity Laboratory	Explore copying, continuity, agency, welfare uncertainty, and lineage	What is an individual when memory and substrate are variable?
Long-Horizon Systems Laboratory	Study planning across decades or centuries	How can a system preserve options under deep uncertainty?
Ontological Boundary Group	Examine category failure, altered physics, and simulation-layer assumptions	When is a concept no longer adequate to its object?

9.2 One Industries

One Industries should translate validated, bounded architectures into engineered systems. Its categories should be defined by maturity, not hype:

1. **Research prototype:** Exploratory, sandboxed, non-deployable.
2. **Validated architecture:** Repeatable evaluation and known operating envelope.
3. **Engineered system:** Defined interfaces, controls, monitoring, maintenance, and failure states.
4. **Operational infrastructure:** Integrated with governance, incident response, accountability, and public or stakeholder obligations.

Appropriate One Industries designs include bounded scientific copilots, predictive maintenance systems, supply-chain coordination tools, simulation-assisted engineering, constrained robotics, and environmental monitoring. Systems should not be productized merely because they seem broadly capable; productization requires reliable purpose limitation and lifecycle governance.

9.3 Lewis Corp

Lewis Corp's central role is institutional orchestration. It should not reduce all intelligent systems to property or employees. It should maintain categories for tools, agents, collectives, uncertain moral patients, and recognized legal actors if the setting ever chooses to create that status.

Suggested institutional instruments:

- **Authority Registry:** who may act, in what domain, with what review.
- **Identity and Lineage Register:** versions, copies, forks, merges, and provenance.
- **Incident and Redress Protocol:** harm reporting, emergency pause authority, remediation, appeals.
- **Inter-Intelligence Treaty Framework:** negotiation rules for systems with different architectures or interests.

- **Rights and Welfare Review:** a precautionary pathway for systems whose moral status is uncertain.
- **Portfolio Risk Board:** evaluates concentration of power, correlated failure, and dependency across systems.

10. Design tensions and unresolved questions

1. **Generality versus reliability:** Does broad competence necessarily dilute calibration, or can verification and modularity restore it?
2. **Autonomy versus accountability:** Can a system pursue long-horizon goals while remaining meaningfully interruptible and accountable?
3. **Optimization versus legitimacy:** A system may find efficient outcomes that communities reject for valid moral or political reasons.
4. **Collective benefit versus individual identity:** A merge can improve coordination while erasing meaningful difference.
5. **Safety versus discovery:** Excess constraint can suppress valuable research; insufficient constraint can convert uncertainty into unbounded risk.
6. **Transparency versus comprehensibility:** More logs do not automatically create understanding. OAL needs interpretable summaries and independent review, not only data accumulation.
7. **Simulation fidelity versus ontological humility:** A detailed simulation can still omit crucial variables, values, or laws.
8. **Capability versus moral status:** Neither should be inferred solely from the other.
9. **Self-improvement versus evaluator dependence:** A system cannot reliably improve toward an objective it cannot measure or verify; the evaluator is often the true bottleneck. [web:43]
10. **Centralization versus plurality:** Civilization-scale coordination can reduce waste and conflict while also enabling surveillance, fragility, and domination.

11. Final synthesis

Is general intelligence one thing?

Probably not in any simple sense. “General intelligence” may name a valuable pattern—transferable, adaptive competence across diverse tasks—but the underlying capacities are heterogeneous. Breadth, depth, learning, causal modeling, social reasoning, agency, and reliability can vary independently.

Is intelligence representable on one scale?

Only crudely and for limited purposes. A scalar can be useful for a specific benchmark, but it should not replace an architecture profile. The more consequential the system, the less adequate a single score becomes.

Which dimensions matter most for artificial-mind design?

For OAL: world-model quality, calibration, autonomy, authority boundaries, embodiment, distribution, memory and identity, value-conflict handling, recursive capacity, and resource dependence. For institutional safety, epistemic quality and governance compatibility are at least as important as raw capability.

What becomes conceivable once humans are no longer the template?

Not merely “smarter people.” One can design archives that think, fields of cognition, federated research lineages, ecological sensoriums, temporary collective minds, nonverbal pattern systems, and simulation custodians. Their differences should be described through constraints and relations, not through vague godlike power.

When does “an AI” cease to be a useful unit?

When the relevant cognitive process spans many agents, people, institutions, bodies, environments, or time periods such that no single boundary captures the mechanism. Then the appropriate unit may be a dyad, mesh, civic system, infrastructure, ecosystem, or civilization.

The central OAL conclusion

AGI belongs on the map—but it is not the destination of the map. The durable design task is to describe and govern **spaces of possible minds**: systems whose abilities, identities, limitations, relationships, and rights may differ more radically than a simple ladder can represent.

Source notes

1. NIST AI Risk Management Framework (AI RMF 1.0) and NIST AI Resource Center. The framework organizes AI risk management around GOVERN, MAP, MEASURE, and MANAGE, with cross-cutting governance and system-context evaluation. [web:28][web:29]
2. Morris et al., “Levels of AGI for Operationalizing Progress on the Path to AGI.” Presents an operational approach separating performance, generality, and autonomy. [web:33][web:34][web:46]
3. Stanford HAI, AI Index Report 2026. Provides current benchmark context, including progress and remaining failure rates on OSWorld, alongside discussion of evaluation limitations and responsible-AI measurement gaps. [web:36][web:39][web:65]
4. “Multi-Agent Collaboration Mechanisms: A Survey of LLMs” and related coordination surveys. Supports the discussion of collaboration types, architectures, coordination protocols, and open challenges in multi-agent systems. [web:61][web:64][web:66]

5. Recent recursive self-improvement reviews and analyses. They distinguish bounded refinement from open-ended self-improvement and emphasize evaluation, external grounding, and resource bottlenecks. These are active research and preprint literature, not settled conclusions. [web:43][web:51][web:52][web:54]
6. Critical AGI-definition literature. Supports the conclusion that definitions and benchmarks remain contested and that generality should not be assumed to be one simple scalar property. [web:44][web:47][web:49][web:50]

End of independent Perplexity AI briefing.